Circle Product Feedback

Why we chose these products

Nanopayments/x402 is the only rail where per-call metered agent payments are economically sane (gasless, sub-cent, batched). Gateway gives the agent a fundable balance with instant receipts. Circle Wallets (DCW EOA) moves key custody out of the agent entirely — the property our whole trust model rests on. USDC-as-gas on Arc collapses faucet/gas UX to one token.

What worked well

• DCW EOA signTypedData signatures are fully ecrecover-compatible with Gateway's x402 settle — the custody-separation design works E2E on Arc Testnet (verified 2026-07-06, transfer 9bd6a149-…-712cdc362382).
• Seller monetization is genuinely 2 lines (createGatewayMiddleware + gateway.require()); per-request dynamic pricing works by binding require(price) at call time — it became our negotiation surface.
• CHAIN_CONFIGS exports eliminated all address/RPC hardcoding.
• Arc deposit finality (~0.5 s) makes the deposit→pay loop feel instant; Wallets contract executions confirm in well under a minute.
• The x402 lifecycle hooks are an excellent extension surface — our entire policy layer lives in one hook.

What could be improved (all found and reproduced during this build)

1. **Nanopayments is EOA-only** (ecrecover on EIP-3009; EIP-1271/SCA unsupported) — easy to miss: Circle's own arc-escrow sample creates accountType SCA wallets, which would silently fail here. Needs a prominent docs callout + runtime hint.
2. **Signature-validity contradiction:** seller quickstart says ≥ 7 days, the SDK error reference says ≥ 3, and the exported GATEWAY_AUTH_VALIDITY_WINDOW_SECONDS constant is undocumented. Pick one number, document the constant.
3. **18-decimal gas vs 6-decimal ERC-20 USDC on Arc** deserves an explicit docs callout — it bites any native-balance math.
4. **Sample apps are heavy references:** arc-escrow needs Supabase + Docker + ngrok + OpenAI before first run; a minimal single-process reference per product would cut onboarding dramatically.
5. **Undocumented header in shipped types:** X-ARC-PRIVATE-MAINNET-ENABLED appears in the SDK's .d.ts, nowhere in docs.
6. **Hook documentation lives off-site** (x402.org, not developers.circle.com) — mirror or link prominently; it's the SDK's best surface.
7. **Broken script in arc-fintech sample:** package.json declares spend:gateway:arc → a file that doesn't exist in the repo.
8. **Wallets ↔ x402 SDK interop wart:** signTypedData requires an explicit EIP712Domain entry in types, but viem-style callers (including BatchEvmScheme's signer interface in Circle's own SDK) omit it. The error — “there is extra data provided in the message (0 < 4)” — names neither the missing type nor the fix. Accept viem-normalized typed data, or return “types.EIP712Domain is required”.
9. **GatewayClient.deposit() requires a raw private key,** so DCW-custodied wallets can't use it. Verified workaround: replicate its two calls (USDC.approve + GatewayWallet.deposit) via createContractExecutionTransaction. A documented DCW-native deposit path is the single change that would most help agentic use cases — Circle-custodied buyers are exactly who agents need.
10. **Deploy vs execute live in different packages:** createContractExecutionTransaction is on @circle-fin/developer-controlled-wallets, but deployContract is only on @circle-fin/smart-contract-platform — a dev working from the Wallets client won't find deploy. Cross-link them.
11. **deployContract rejects an empty constructorParameters: []** with a generic 400 that names no field; omitting it works. Accept [] or name the bad param.
12. **Gas Station (SCA-only) and Nanopayments/x402 (EOA-only) are mutually exclusive for one wallet** — agentic builders want both; emit a runtime hint on the mismatch.

13. **No single-call DCW-native CCTP helper:** cross-chain top-ups work by chaining approve → depositForBurn → IRIS attestation → receiveMessage via contract-execution, but need a funded DCW wallet per source chain. A DCW-native CCTP method would unlock autonomous cross-chain treasury for agents.

Recommendations

Ship a first-class "agent buyer" recipe: DCW EOA wallet + Gateway deposit via contract execution + x402 client with a policy hook — the five pieces exist today but must be discovered across four doc sites and two SDKs' type definitions. This project is, in effect, that recipe; we'd be glad if it were useful as a reference.
